Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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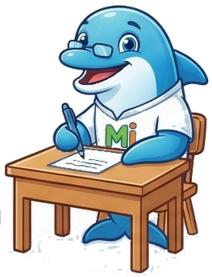
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Code the pattern**	6-7	8-9	10 and 12	11	☐
2	**Prime hunters**	Not in this proof edition				☐
3	**Below zero**	Not in this proof edition				☐
4	**Algebra machines**	Not in this proof edition				☐
5	**Every possible way**	Not in this proof edition				☐
6	**Fair or unfair?**	Not in this proof edition				☐
7	**Turtle geometry**	Not in this proof edition				☐
8	**Secret codes**	Not in this proof edition				☐
9	**Data detectives**	Not in this proof edition				☐
10	**Showcase challenge**	Not in this proof edition				☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Sequences and the nth term

My goal. I can find the rule for the nth term of a pattern and use it for the 100th term. I can test my rule with a loop.

A rule for the nth term jumps straight to any term, such as the 100th. The difference tells you which times table to compare with. A loop in code adds the difference again and again, so you can test your rule.

Worked example

Find the rule for the nth term of 9, 14, 19, 24, … Then find the 100th term.

n	1	2	3	4
5 × n	5	10	15	20
term	9	14	19	24

1. The difference is 5, so compare each term with the 5 times table.
2. Each term is 4 more than the number above it. The rule is 5n + 4.
3. Check the 4th term: 5 × 4 + 4 = 24. It matches.
4. The 100th term is 5 × 100 + 4 = 504.
5. Test it in code: set term to 9, then repeat 99 times: change term by 5. The 100th term is 99 jumps after the 1st term.

Answer: The rule is 5n + 4. The 100th term is 504.

How to find the rule for the nth term

1. Find the difference between neighbouring terms.
2. Write that times table under n: 5, 10, 15, …
3. Find how much to add or take away to reach each term.
4. Check the rule on 2 terms. Then use it for any term.

Your turn

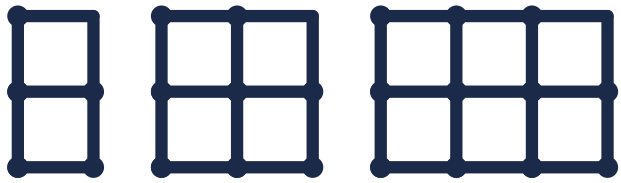
Answer at the back of the book.

Try **A sequence starts 3, 10, 17, 24, … The rule for the nth term is ☐ × n – ☐. The 100th term is ☐.**

Start here

Do these first. Take your time.

1 **Matchsticks make 2 rows of squares. Each new pattern adds one more column of 2 squares.**



Pattern 1 Pattern 2 Pattern 3

(a) Pattern 3 has ☐ sticks. Each new column adds ☐ sticks.

(b) The rule: sticks = ☐ × n + ☐

(c) Pattern 40 has ☐ sticks.

(d) Pattern ☐ has 92 sticks.

2 **For each sequence, fill in the difference, the rule for the nth term and the 100th term.**

Sequence	Difference	nth term	100th term
(a) 8, 14, 20, 26, ...			
(b) 3, 8, 13, 18, ...			

3 **A sequence has the rule 5n + 2. Its first terms are 7, 12, 17, 22, ...**

(a) Is 157 a term? Yes No If yes, it is term number ☐.

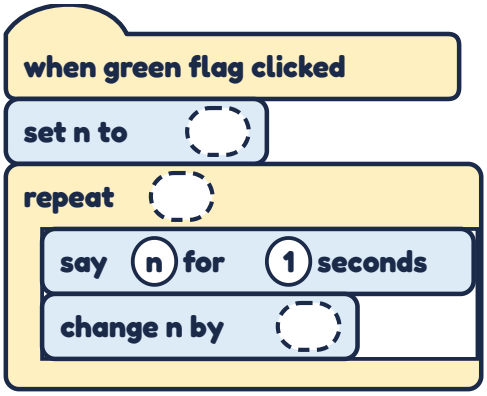
(b) Is 205 a term? Yes No

(c) The 60th term is ☐.

Start here

Keep going. Check your work.

5 Complete the code so that the sprite says the first 15 terms of 11, 18, 25, 32, ...



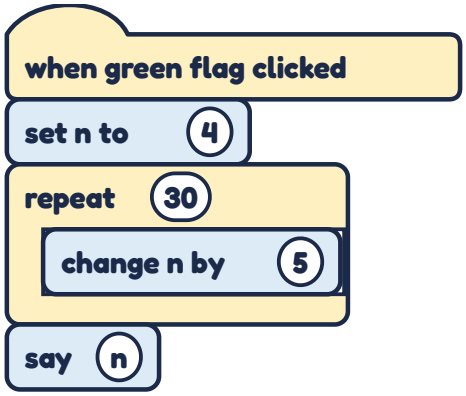
(a) set n to ☐

(b) repeat ☐

(c) change n by ☐

(d) The last number it says is ☐.

6 Siti's code should say the 30th term of 4, 9, 14, 19, ...



(a) Her code says ☐.

(b) The 30th term really is ☐.

(c) To fix it, change repeat 30 to repeat ☐.

Start here

Nearly there. You can do it.

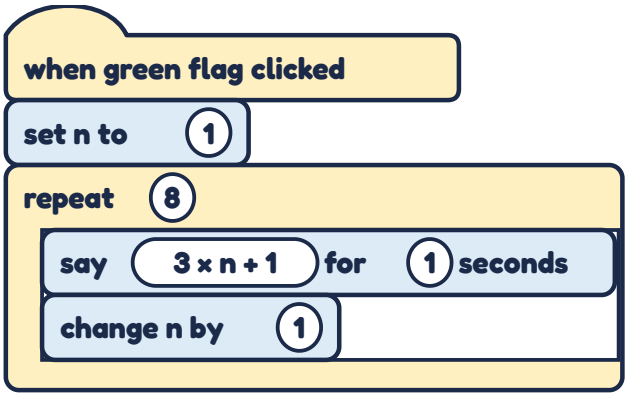
9 **A loop starts at the 1st term of a sequence. Each repeat moves on to the next term.**

(a) To reach the 60th term, repeat ☐ times.

(b) After repeat 40, the sprite is at term number ☐.

(c) To go from the 8th term to the 25th term, repeat ☐ times.

10 **This code uses n as a counter and says a rule. Predict what it says.**



(a) The first 3 numbers it says are ☐, ☐, ☐.

(b) The last number it says is ☐.

(c) To say 5, 9, 13, 17, ... change the rule to ☐ × n + ☐.

11 **Use the rules to find terms.**

(a) The 20th term of 6n + 1 is ☐.

(b) The 30th term of 9n – 4 is ☐.

(c) In 6n + 1, the number 61 is term number ☐.

(d) In 9n – 4, the number 158 is term number ☐.

Think harder

Hints are upside down. Try first!

1 **A sequence goes down by the same amount each time: 58, 53, 48, 43, …**

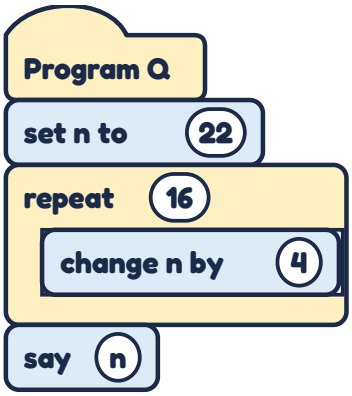
(a) The rule for the nth term is ☐ − 5n.

(b) The 10th term is ☐.

(c) The last term above 0 is term number ☐.

Hint. The terms go down by 5. Take 5 × n away from a start number.

2 **Program P starts at the 1st term of 6, 10, 14, 18, … Program Q starts at the 5th term of the same sequence. Both change by 4 each repeat.**



(a) Q ends at term number ☐.

(b) To end at the same term, P must repeat ☐ times.

(c) Both programs say ☐.

Hint. Count how many terms Q moves on. P starts 4 terms earlier.

3 **The nth term of a sequence is 9n + 4.**

(a) The term closest to 500 is ☐.

(b) It is term number ☐.

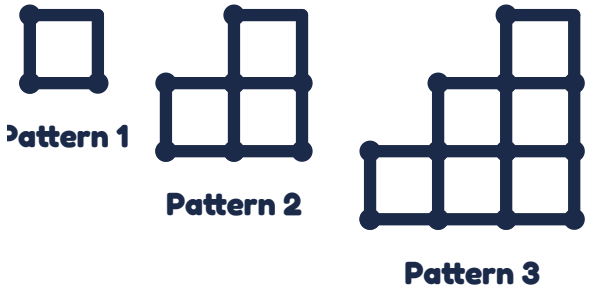
(c) The number of terms that are less than 300 is ☐.

Hint. Try n = 50 first. Then move up or down until you are close.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 6: Second differences.**
Matchsticks make staircases. Staircase 1 has 1 square, staircase 2 has 3 squares and staircase 3 has 6 squares.



(a) Count the sticks. Staircase 2 has ☐ and staircase 3 has ☐.

(b) The extra sticks go up by the same amount each time. Staircase 4 has ☐ sticks and staircase 5 has ☐.

(c) Staircase 10 has ☐ sticks.

Hint. Find how many extra sticks each staircase adds. Look for a pattern.

2 **A sprite says these numbers in order: 3, 20, 7, 25, 11, 30, 15, 35, ...**

(a) Places 1, 3, 5, ... go up by ☐. Places 2, 4, 6, ... go up by ☐.

(b) The number in place 15 is ☐.

(c) The number in place 20 is ☐.

(d) The number 60 is in place ☐.

Hint. Split it into two sequences. Which one is place 15 in?

3 **A sequence has the rule 6n – 2. Plan a program that starts at the 1st term and says the 40th term.**

(a) The first 4 terms are ☐, ☐, ☐, ☐.

(b) The 40th term is ☐.

(c) set n to ☐, then repeat ☐ [change n by ☐], then say n.

Hint. How many jumps are there from the 1st term to the 40th term?

✔ Check yourself

No hints. Try each one on your own.

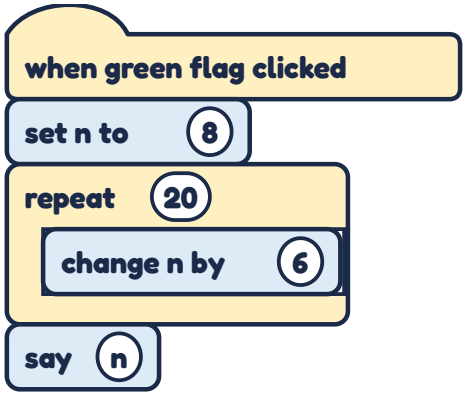
1 Find the rule for 9, 16, 23, 30, … The rule is ☐ × n + ☐. The 100th term is ☐.

Working: _____

2 Is 125 a term of 7, 12, 17, 22, …? Yes No

Working: _____

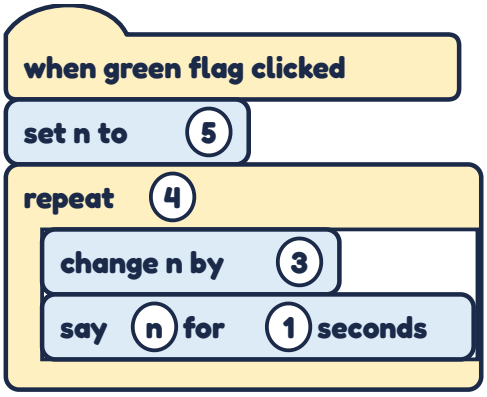
3 Predict what the code says.



(a) It says ☐.

(b) That is term number ☐ of 8, 14, 20, …

4 Mei wants the first 4 terms of 5, 8, 11, 14. What does her code say?



(a) It says ☐, ☐, ☐, ☐.

(b) The number it never says is ☐

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 6.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Code the pattern

Your turn
page 6

Answer: 7, 4, 696
Why: The difference is 7, so compare with 7, 14, 21, 28. Each term is 4 less than the number under it. The rule is 7n − 4. The 100th term is 7 × 100 − 4 = 696.

START HERE

1
page 7

Answer: (a) 17, 5 (b) 5, 2 (c) 202 (d) 18
Why: Count: pattern 1 has 7 sticks, pattern 2 has 12, pattern 3 has 17. Each new column adds 5 sticks, so the rule is 5n + 2 (5 × 1 + 2 = 7). Pattern 40: 5 × 40 + 2 = 202. For 92 sticks: 5n + 2 = 92, so 5n = 90 and n = 18.

2
page 7

Answer: (a) 6, 6n + 2, 602 (b) 5, 5n − 2, 498
Why: (a) The difference is 6. Compare with 6, 12, 18, 24: each term is 2 more, so 6n + 2 and 6 × 100 + 2 = 602. (b) The difference is 5. Each term is 2 less than 5, 10, 15, 20, so 5n − 2 and 5 × 100 − 2 = 498.

3
page 7

Answer: (a) Yes, term 31 (b) No (c) 302
Why: (a) 157 − 2 = 155 = 5 × 31, so 157 is the 31st term. (b) 205 − 2 = 203 is not a multiple of 5. The terms end in 2 or 7, so 205 is not one. (c) 5 × 60 + 2 = 302.

4
page 7

Answer: (a) 21 (b) 27 (c) 297
Why: Loop 1: it says 3, then x becomes 9. Loop 2: says 9, x becomes 15. Loop 3: says 15, x becomes 21. Loop 4: says 21, x becomes 27. The last number said is 21 but x ends as 27. With repeat 50 it says the 1st term, then 49 jumps later it says 3 + 49 × 6 = 297.

5
page 8

Answer: (a) 11 (b) 15 (c) 7 (d) 109
Why: The 1st term is 11, so set n to 11. Each jump adds 7. The sprite says a term, then changes n, and 15 terms need 15 says: repeat 15. The last number said is the 15th term: 7 × 15 + 4 = 109.

6
page 8

Answer: (a) 154 (b) 149 (c) 29
Why: The loop adds 5 thirty times: 4 + 5 × 30 = 154. That is the 31st term. The 30th term is 5 × 30 − 1 = 149. From the 1st to the 30th term there are only 29 jumps, so use repeat 29.

7
page 8

Answer: (a) 36 (b) 7, 8 (c) 218 (d) 28
Why: Each day adds $7, so the rule is 7n + 8 (7 × 1 + 8 = 15). (a) Day 4: 7 × 4 + 8 = 36. (c) Day 30: 7 × 30 + 8 = 218. (d) 7n + 8 is more than 200 when 7n is more than 192. 7 × 27 = 189 is too small, 7 × 28 = 196 works: Day 28 (he has $204).

8
page 8

Answer: (a) 5n − 1 (b) 3n + 7 (c) 8n − 5
Why: Put n = 1 and n = 2 into each rule. (a) 5 × 1 − 1 = 4 and 5 × 2 − 1 = 9. (b) 3 × 1 + 7 = 10 and 3 × 2 + 7 = 13. (c) 8 × 1 − 5 = 3 and 8 × 2 − 5 = 11. The other rules fail on the 1st term.

9
page 9

Answer: (a) 59 (b) 41 (c) 17
Why: The 1st term needs 0 repeats and each repeat adds 1 to the term number. (a) 60 − 1 = 59 repeats. (b) 1 + 40 = the 41st term. (c) 25 − 8 = 17 repeats. The repeats count the jumps, not the terms.

10
page 9

Answer: (a) 4, 7, 10 (b) 25 (c) 4, 1
Why: (a) n = 1, 2, 3 give 3 × 1 + 1 = 4, 7, 10. (b) The 8th number is 3 × 8 + 1 = 25. (c) 5, 9, 13 go up by 4, so use 4 × n + 1: 4 × 1 + 1 = 5.

11
page 9

Answer: (a) 121 (b) 266 (c) 10 (d) 18
Why: (a) 6 × 20 + 1 = 121. (b) 9 × 30 − 4 = 266. (c) 6n + 1 = 61 gives 6n = 60, so n = 10. (d) 9n − 4 = 158 gives 9n = 162, so n = 18.

12
page 9

Answer: (a) 5 (b) 6 (c) 5 (d) 6, 1
Why: From the 4th to the 9th term is 9 − 4 = 5 jumps. The terms rose by 30, so each jump is 30 ÷ 5 = 6. Going back 3 jumps from the 4th term: 23 − 18 = 5. The rule is 6n − 1 (6 × 4 − 1 = 23).

13
page 9

Answer: (a) No (b) 6, 2 (c) 6, 4
Why: The rule 6n gives 6 for the 1st term, but the 1st term is 2. Each term is 4 less than 6n: 6 − 4 = 2, 12 − 4 = 8. The difference only tells you the times table. You must still compare with the terms.

THINK HARDER

1
page 10

Answer: (a) 63 (b) 13 (c) 12
Why: (a) The terms go down by 5, so take 5n from a number. 5 × 1 = 5 and 58 + 5 = 63, so the rule is 63 − 5n. (b) 63 − 50 = 13. (c) 63 − 5n stays above 0 while 5n is less than 63. 5 × 12 = 60 gives 3 and 5 × 13 = 65 would be below 0, so term 12.

2
page 10

Answer: (a) 21 (b) 20 (c) 86
Why: The 5th term is 4 × 5 + 2 = 22. Q adds 4 sixteen times: 22 + 64 = 86. Q has moved 16 terms on, so it ends at term 5 + 16 = 21. P starts at the 1st term, so it needs 21 − 1 = 20 repeats. Check: 6 + 20 × 4 = 86.

3
page 10

Answer: (a) 499 (b) 55 (c) 32
Why: (a) 9n + 4 is near 500 when n is near 55: 9 × 55 + 4 = 499 and 9 × 56 + 4 = 508. 499 is only 1 away from 500. (b) It is the 55th term. (c) 9n + 4 is less than 300 while 9n is less than 296. 9 × 32 = 288 works and 9 × 33 = 297 does not, so 32 terms.

GO FURTHER

1
page 11

Answer: (a) 10, 18 (b) 28, 40 (c) 130
Why: The extra sticks each time are 6 (4 to 10), then 8 (10 to 18). They go up by 2 each time, so the next extras are 10 and 12: 18 + 10 = 28 and 28 + 12 = 40. For staircase 10 add the extras 6, 8, ..., 22 to 4: 4 + 126 = 130. (The rule is n × n + 3n, a rule that is not a times table plus a number.)

2 page 11

Answer: (a) 4, 5 (b) 31 (c) 65 (d) 18
Why: Places 1, 3, 5, … say 3, 7, 11, 15: they go up by 4. Places 2, 4, 6, … say 20, 25, 30, 35: they go up by 5. Place 15 is the 8th of the first kind: 3 + 7 × 4 = 31. Place 20 is the 10th of the second kind: 20 + 9 × 5 = 65. 60 is in the second kind: 20 + 8 × 5 = 60 is its 9th number, which is place 18.

3 page 11

Answer: (a) 4, 10, 16, 22 (b) 238 (c) 4, 39, 6
Why: n = 1, 2, 3, 4 give 4, 10, 16, 22. The 40th term is 6 × 40 − 2 = 238. The program starts at the 1st term, 4, and jumps 6 each time. From the 1st to the 40th term there are 39 jumps, so: set n to 4, repeat 39 [change n by 6], say n.

4 page 11

Answer: (a) 14 (b) 105
Why: t after each repeat: 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105. The repeat stops the first time t is above 100, which is after repeat 14, so k is 14 and t is 105. (These are the triangle numbers 1, 3, 6, 10, … : the extra amount is 1 more each time.)

CHECK YOURSELF

1 page 12

Answer: 7, 2, 702
Why: The difference is 7. Each term is 2 more than 7, 14, 21, 28, so the rule is 7n + 2. The 100th term is 7 × 100 + 2 = 702.

2 page 12

Answer: No
Why: The rule is 5n + 2. 125 − 2 = 123 is not a multiple of 5. The terms end in 2 or 7, so 125 is not a term.

3 page 12

Answer: (a) 128 (b) 21
Why: 8 + 20 × 6 = 128. From the 1st term to the 21st term there are 20 jumps, so 128 is the 21st term. Check: 6 × 21 + 2 = 128.

4 page 12

Answer: (a) 8, 11, 14, 17 (b) 5
Why: Each loop changes n first, so it says 8, 11, 14, 17. The 1st term 5 is never said, and 17 is one term too far. To fix it, swap the 2 blocks inside the repeat.